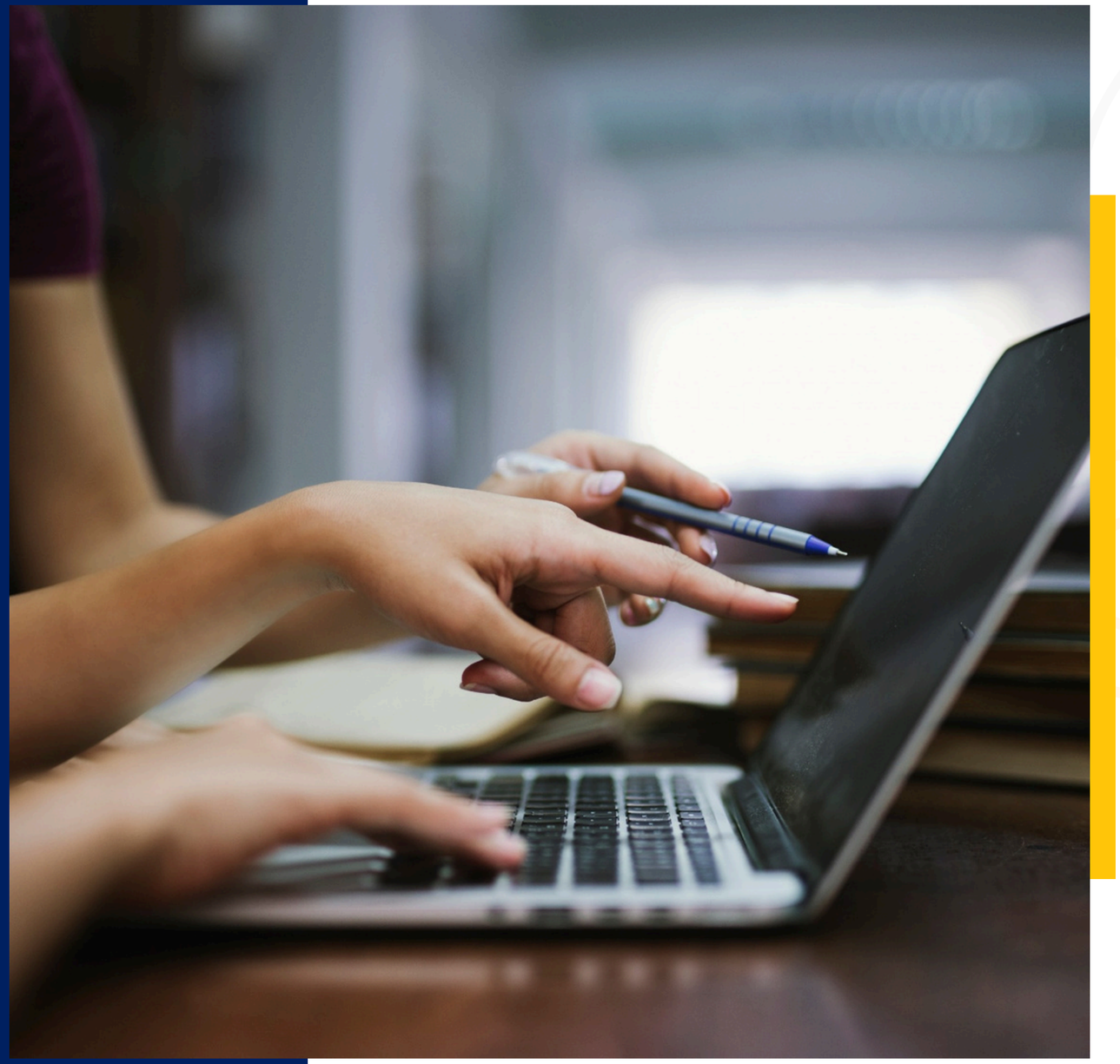

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Cyber Sentinel

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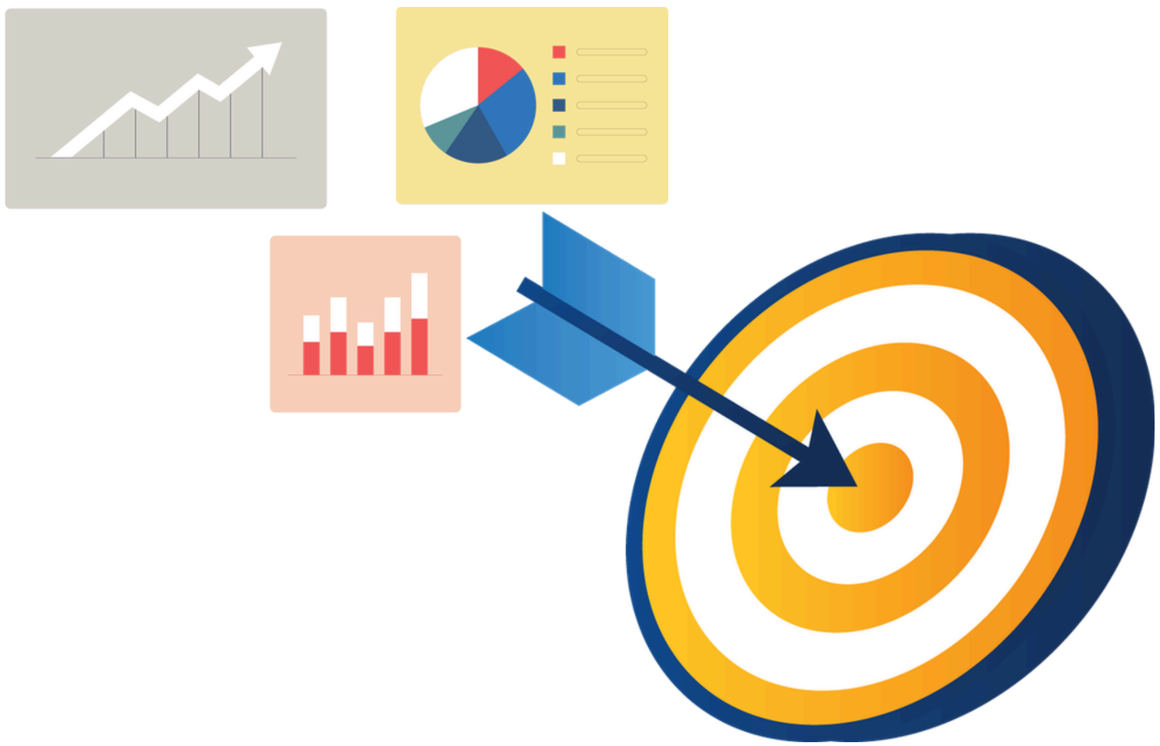
Project Objectives

- Problem Statement
- Project Overview – Introduction
- End Users
- Wow Factor in Project
- Modelling/Block Diagram/Flow of Project
- Result/outcomes
- Conclusion
- Future Perspective



Source

https://www.freepik.com/

Problem Statement

- Organizations face increasing cyber threats that traditional security systems fail to detect in real-time. Manual threat analysis is slow, error-prone, and cannot scale with growing network traffic. There's a critical need for an automated, intelligent system that can detect anomalies, classify threats, and suggest immediate responses. Such a system also helps reduce human workload, improves accuracy, and strengthens overall resilience against fast-moving attacks.

Project overview - Introduction

CyberSentinel is an **AI-powered Security Operations Center (SOC) dashboard** that combines:

- **Machine Learning**: Uses Isolation Forest algorithm to detect network anomalies
- **Agentic AI**: Intelligent rule-based decision making for threat classification
- **Real-time Monitoring**: Live attack feed visualization
- **Automated Response**: Recommends actions like "Block IP" or "Isolate System"

The system analyzes network logs including:

- Packet count
- Failed login attempts
- Session duration

End User

- Security Analysts - Monitor threats and investigate alerts
- SOC Teams - Get automated threat intelligence
- IT Administrators - Receive actionable response recommendations
- Small to Medium Businesses - Affordable security monitoring solution
- Educational Institutions - Learn about AI in cybersecurity

Wow factor

1.Dual-AI Architecture

Machine Learning (Isolation Forest) + Agentic AI (Rule-based Reasoning) in one pipeline. ML detects unknown anomalies while Agentic AI provides intelligent, context-aware decisions.

2. Explainable AI with Automated Response

Each detection includes a "why_risky" explanation and recommends specific actions like "Block IP & Isolate System" - complete transparency with actionable outputs.

3. Real-time Attack Simulation

Live attack feed auto-refreshes every 5 seconds simulating DDoS, Brute Force, Port Scans, and Malware Traffic - creates immersive SOC experience without real attack infrastructure.

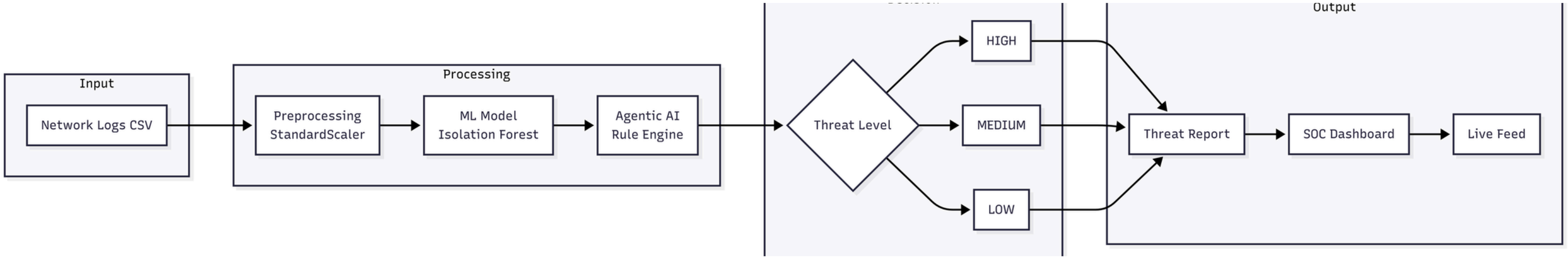
4. Multi-threat Intelligence

Identifies 4+ specific attack types beyond binary classification - provides attack-specific context for faster, more accurate incident response.

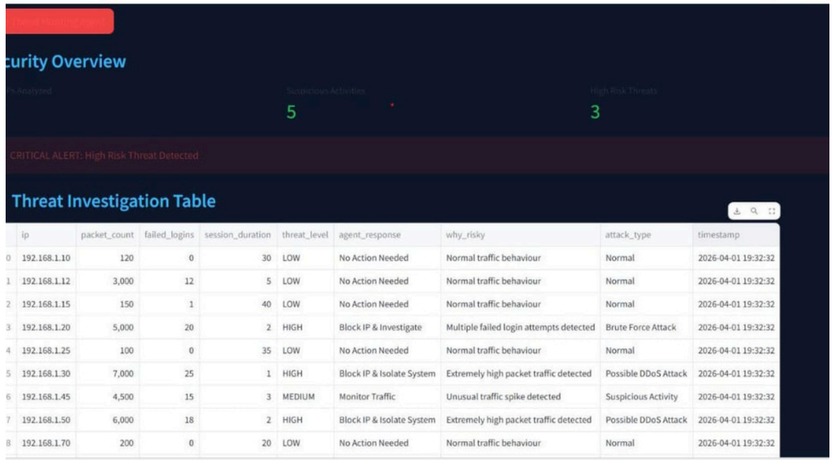
5. Zero-configuration Deployment

Works out-of-the-box with automatic directory creation, no database setup, no training data required - deploy within minutes with professional-grade SOC dashboard.

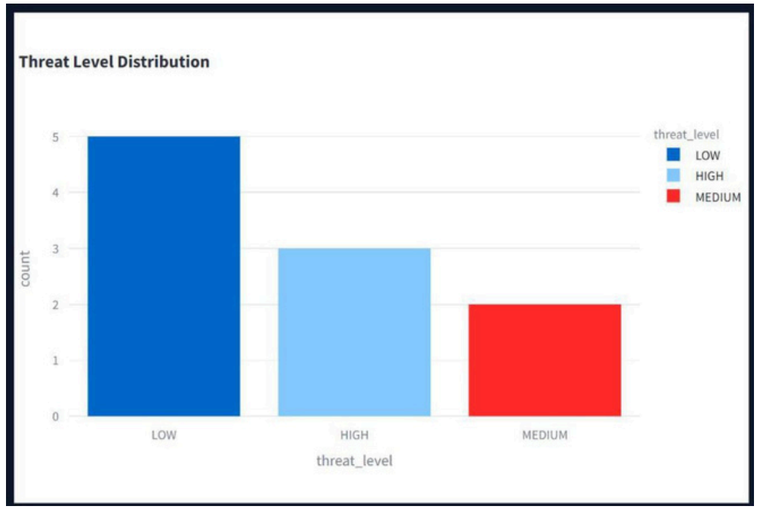
Modelling

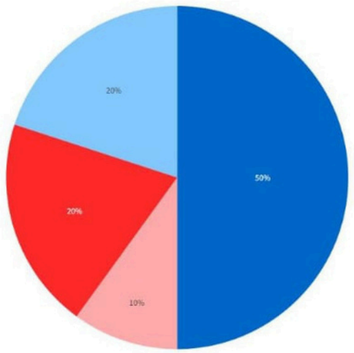

Input
Network Logs CSV
Processing
Preprocessing StandardScaler
ML Model Isolation Forest
Agentic AI Rule Engine
Threat Level
HIGH
MEDIUM
LOW
Output
Threat Report
SOC Dashboard
Live Feed

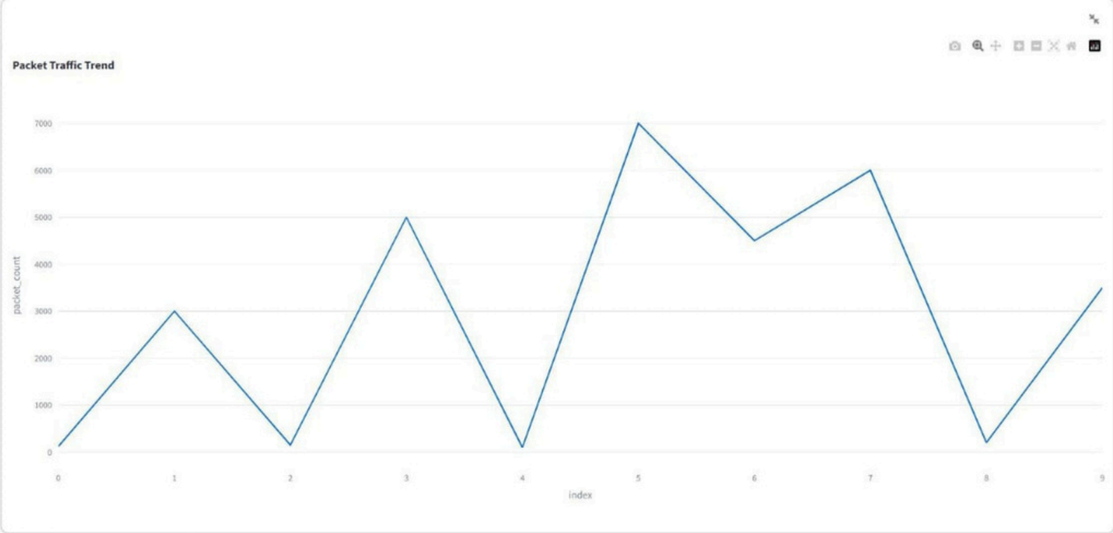
Result / Outcomes



	ip	packet_count	failed_logins	session_duration	threat_level	agent_response	why_risky	attack_type	timestamp
0	192.168.1.10	120	0	30	LOW	No Action Needed	Normal traffic behaviour	Normal	2026-04-01 19:32:32
1	192.168.1.12	3,000	12	5	LOW	No Action Needed	Normal traffic behaviour	Normal	2026-04-01 19:32:32
2	192.168.1.15	150	1	40	LOW	No Action Needed	Normal traffic behaviour	Normal	2026-04-01 19:32:32
3	192.168.1.20	5,000	20	2	HIGH	Block IP & Investigate	Multiple failed login attempts detected	Brute Force Attack	2026-04-01 19:32:32
4	192.168.1.25	100	0	35	LOW	No Action Needed	Normal traffic behaviour	Normal	2026-04-01 19:32:32
5	192.168.1.30	7,000	25	1	HIGH	Block IP & Isolate System	Extremely high packet traffic detected	Possible DDoS Attack	2026-04-01 19:32:32
6	192.168.1.45	4,500	15	3	MEDIUM	Monitor Traffic	Unusual traffic spike detected	Suspicious Activity	2026-04-01 19:32:32
7	192.168.1.50	6,000	18	2	HIGH	Block IP & Isolate System	Extremely high packet traffic detected	Possible DDoS Attack	2026-04-01 19:32:32
8	192.168.1.70	200	0	20	LOW	No Action Needed	Normal traffic behaviour	Normal	2026-04-01 19:32:32







Conclusion

CyberSentinel successfully demonstrates:

- Effective anomaly detection using Isolation Forest ML algorithm
- Intelligent threat response through Agentic AI decision-making
- User-friendly SOC dashboard for security monitoring
- Scalable architecture that can process multiple network logs
- Explainable outputs helping analysts understand threat reasoning
- The system bridges the gap between ML-based detection and actionable security responses, making it valuable for modern SOC operations.

Source

https://www.freepik.com/

Future Perspective

1. Real-time Packet Capture Integration

Integrate with live network traffic (pcap files) and packet sniffing tools like Wireshark to analyze real-time data streams instead of static CSV files.

2. Automated Threat Mitigation

Connect with firewalls, routers, and IDS/IPS systems to automatically block malicious IPs and isolate compromised systems without manual intervention.

3. Deep Learning Models

Implement advanced neural networks like LSTM for time-series analysis and CNN for pattern recognition to detect sophisticated zero-day attacks.

4. Mobile SOC Application

Develop a mobile app for iOS and Android allowing security teams to monitor threats, receive push notifications, and respond to incidents from anywhere.

5. Threat Intelligence Integration

Connect with external threat intelligence feeds (VirusTotal, AlienVault, IBM X-Force) to enrich detection with global threat data and improve accuracy.

Thank You